

ABRIAN ABIR, M.S., E.I.T.

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EDUCATION

The University of Tennessee

Master of Science with Mechanical Engineering (*GPA: 4.00/4.00*)

Bachelor of Science in Mechanical Engineering; Minor in Computer Science (*GPA: 3.99/4.00*)

Knoxville TN

May 2026

May 2024

LICENSES & CERTIFICATIONS

Engineer-in-Training (EIT), TN Department of Commerce and Insurance

Oct 15, 2025

Certified SOLIDWORKS Associate (CSWA), Dassault Systèmes

Nov 27, 2023

EXPERIENCE

Oak Ridge National Laboratory

Oak Ridge, TN

Applications Engineer for Advanced Manufacturing

Sep 2026 - Present

- Design, install, and calibrate advanced image-based and multi-sensor systems to ensure data integrity for machine learning applications.
- Develop custom data acquisition software and novel sensing techniques to upgrade and maintain optical/sensing infrastructure.
- Support cross-functional R&D projects for Manufacturing Demonstration Facility (MDF) aligned with core ORNL mission objectives.

Mechanical Systems Engineering Intern (SULI) – Fusion Research

May 2024 - Aug 2024

- Designed adaptive topology optimization framework to generate quantitative Figures of Merit (FoM), improving cooling performance.
- Automated multi-physics workflows by integrating HEAT and ANSYS Fluent using pyAnsys and PyGAD to reduce design iteration.
- Validated structural integrity of WEST tokamak probes and PFCs via coupled CFD/FEA under high heat-flux and transient EM loads.

DENSO Manufacturing Tennessee, Inc.

Maryville, TN

Process Engineer I – Automation & Controls

Jun 2026 - Sep 2026

- Lead new product introduction, process planning, trials, and production capacity studies including line layouts and cost analysis.
- Manage new equipment specification, procurement, installation, and automation projects (including vision-guided robotics).
- Drive process optimization, root cause analysis, and IATF documentation for production improvements (PFMEA, PCS, Process Flows).

University of Tennessee

Knoxville, TN

Graduate Researcher and Teaching Instructor

Aug 2024 - Jun 2026

- Engineered Python dissimilarity metrics for in-situ AFSD monitoring to enable automated defect detection from raw telemetry.
- Implemented thermal imaging for the live tracking of temperature during microwave power characterization for enclosed components.
- Instructed 30+ students in Mechanical Engineering Laboratory (ME 449), focused on data interpretation and technical documentation.

Undergraduate Researcher and Teaching Assistant

Nov 2020 - Aug 2024

- Designed & fabricated automated 3-point bend tester using Arduino-based custom software to characterize novel composite materials.
- Automated elastomeric 3D print buckling and deflection analysis using Python/OpenCV, reducing manual processing time.
- Mentored 60+ students in MATLAB and Engineering Fundamentals with faculty, managing project evaluations and weekly instruction.

Research Assistant Internship

Aug 2019 - May 2020

- Created Python visualization tools to standardize calculation of Young's modulus, UTS, and yield strength from raw MTS test data.
- Developed Python machine-vision applications (OpenCV) for automated inspection and visual tracking of thermoset 3D printing.

Eastman Chemical Company

Kingsport, TN

Project & Controls Lead – Capstone Project

Aug 2023 - May 2024

- Led full-lifecycle development of film separator prototype, managing cross-functional team coordination & system-level requirements.
- Developed C++ closed-loop control software for dynamic motor operation, utilizing real-time sensor feedback to maintain film tension.
- Authored technical data package, from assembly drawings to specifications, to support the transition from prototype to production.

GE Vernova Hitachi Nuclear Energy

Wilmington, NC

Nuclear Fuel Performance and Design Intern

May 2023 - Aug 2023

- Constructed a Python-based comprehensive database for 200+ BWR fuel units to identify failure signatures across five reactor sites.
- Conducted root cause analysis (RCA) and failure assessments using thermal-mechanical models to identify high-risk reactor rod cells.
- Presented critical failure findings to thermal-mechanical team, enabling data-driven preventative inspections and NQA-1 compliance.

FIRST Robotics (FRC) - Team 6517

Knoxville, TN

Robotics Team Lead

Aug 2016 - May 2020

- Directed design and manufacture of competitive robot components using SOLIDWORKS, Fusion 360, CNC milling, and 3D printing.
- Led the curriculum development for Computer Science and Robotics while mentoring 30+ students in robot design, C++, and Python.

PUBLICATIONS

Additive Manufacturing Letters | Additive friction stir deposition of multi-layer aluminum-silicon carbide metal matrix composites

Additive Manufacturing | Structural stability of thin overhanging walls extrusion additive manufacturing of thermoset-based ink

Additive Manufacturing | Linking thermoset ink rheology to the stability of 3D-printed structures

Journal of Manufacturing Processes | A Dissimilarity Index for Monitoring Process in Additive Friction Stir Deposition

TECHNICAL SKILLS

Simulation & Analysis: ANSYS Fluent, Mechanical, CFD, FEA, pyAnsys, topology optimization, heat transfer

Programming & Controls: Python (NumPy, OpenCV, TensorFlow), C++, MATLAB, Arduino, Closed-Loop Control

Design & Manufacturing: SOLIDWORKS (CSWA), GD&T, Additive Manufacturing (AFSD, FDM, SLA), CNC Milling

AWARDS AND COMMUNITY SERVICE

Academic: Summa Cum Laude, 1794 Honors Scholar, Outstanding Senior & Junior in MAE, Order of the Engineer

Service: Keep Knoxville Beautiful, Habitat for Humanity